

The Utility Problem

As electricity gains a more significant share of overall energy consumption through electrification of transportation, heating and cooking, the electricity distribution grid experiences more strain. One of the specific culprits is a rapid rise in (Electric Vehicle) EVs and the consequent need for EV charging. In the distribution grid, each of the grid elements has specific properties. For instance, transformers have a certain “capacity” (or kVA rating) which dictates the maximum electrical load that they can support at any given time. This, along with the existing electrical load, restricts how many new EV chargers (as well as other loads) can be installed “downstream” of a given transformer without causing it to “overload”. Load is typically not directly measured at the transformer but at each of the consumer locations downstream of it (often modelled via meters). Therefore existing transformer load needs to be aggregated from all these loads. Moreover, it is a dynamic quantity modelled as a time series. During “peak periods” (when the load is high), there will be less available capacity for additional EV chargers, whereas during “off-peak periods,” more EV chargers could be supported. For the given use case, it is necessary to answer the question: “how many new EV chargers could be accommodated?” for both of these scenarios.

Results from such analysis provide valuable insights to multiple departments in the utility. Grid planners will be better able to understand the need for an increased load capacity in specific areas of the service area. Asset managers will also benefit from such information as they can manage bottlenecks on the transformer asset side. Last but not least, the customer account management department will get information about where additional permits for EV chargers can be issued without the need for upgrading the grid infrastructure. All this can help save significant investment costs by optimizing the existing grid.

The Data Solutions and Methods

For conducting such an analysis, it is necessary to have several datasets ready. These datasets are as follows:

1. Transformer asset information (properties of the transformer), specifically the transformer capacity - kVA rating.
2. Knowledge of the connected assets (loads) to the transformer
3. Knowledge about the dynamic load of the connected assets or load at the transformer (load time-series data)
4. Knowledge about the most common EV charger types along with their active charging power

In addition, the above-mentioned datasets should be synchronized and aligned in time and space to create a unified data layer. Awesense Digital Energy Platform offers these capabilities. Awesense Platform’s unique linkage of geospatial data (GIS) with time-series data enables this type of analysis. With data populated in the Awesense Platform, the Energy Data Model’s (EDM)

database views and functions will accelerate the building of the Transformer Capacity Analysis for EV Charging.

To get the necessary datasets for the Transformer Capacity Analysis for EV Charging, the Awesense EDM offers multiple methods:

1. Transformer kVA rating:
 - a. This comes from the core geospatial dataset for Awesense Digital Energy Platform - the GIS data. GIS data often includes information about transformer kVA rating. This information is modelled in EDM as a transformer.rating_kva.
2. Connected assets (loads) to the transformer:
 - a. This can be obtained by using a special purpose database function called grid tracing. This function will use GIS connectivity to retrieve all connected assets in the required direction (upstream, downstream, or same voltage).
 - b. Another method is to use the EDM element property called the parent element, which should yield similar results (since the parent element information is the hierarchical data source, discrepancies might exist).
3. Load time-series:
 - a. Some utilities have measurement devices located at the transformer level. This data can be integrated into the Awesense Digital Energy Platform, linked to the transformer asset and easily obtained by functional call.
 - b. In the cases when the load time series are not available on the transformer level, the load can be inferred from the downstream assets' load time series. This can be done using the list of connected assets (loads) and summing up their load-time series. Awesense EDM offers simplified functions for such types of calls.
4. Most common EV charger:
 - a. This can be obtained by analysis of the GIS data elements called EV Charger in EDM or utilities might have their own list of preferred/most-common EV chargers against which they would like to conduct the analysis. The active charging power is a property of the EV charger element in the EDM data model.

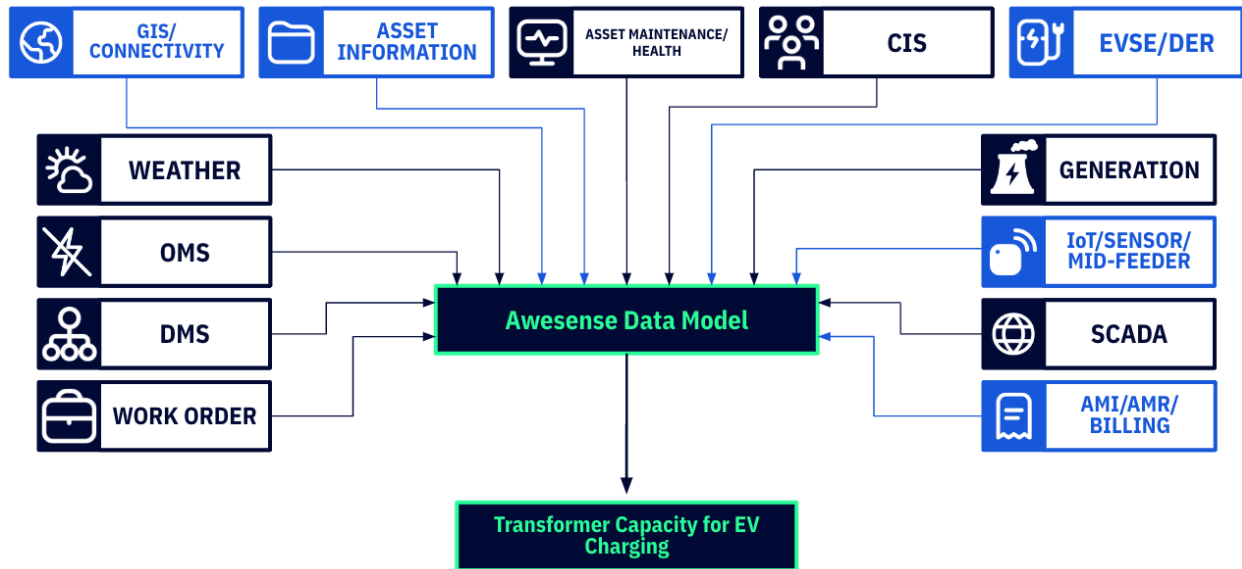
To put this all together, the result of this analysis is a variable amount of EV chargers over time which can be supported by the free capacity of the serving/supplying transformer(s). Such results can be reported on a one-by-one basis or on a grid-wide level (all depending on the specifics of the given implementation).

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.

Transformer Capacity Analysis for EV Charging

[UC04-01] - Awesense Digital Energy Platform Use Case



Available Use Case Implementations:

Users of the Awesense Energy Data Platform can access this use case by the following means:

1. GitHub link to the Jupyter notebook can be provided upon request.